

Last login: Thu Jun 18 11:00:24 on ttys000

```
stephenrajva@Stephens-MacBook-Air-4 ~ % mkdir -p "/Users/stephenrajva/Downloads/SST_Stimulator/Communication_Pattern/HalfShift_experiment"
stephenrajva@Stephens-MacBook-Air-4 ~ % head -100 ~/Downloads/HPC-Upload\ 2/mpi-patterns/mpi_halfshift.c
```

```
#include <stdio.h>
#include <stdlib.h>
#include <dlfcn.h>
#include <mpi.h>

int main(int argc, char *argv[]) {
    int rank, size, N;
    int *data = NULL;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // --- ADDED THIS BLOCK TO GET MAC BASE ADDRESS ---
    if (rank == 0) {
        Dl_info info;
        dladdr((void*)main, &info);
        printf("\n=== MAC BASE ADDRESS: %p ===\n\n", info.dli_fbase);
    }
    // -----

    if (argc != 2) {
        if (rank == 0) printf("Usage: mpirun -np <processes> %s <N>\n", argv[0]);
        MPI_Finalize();
        return 1;
    }

    N = atoi(argv[1]);
    if (size % 2 != 0) {
        if (rank == 0) printf("Error: Requires an EVEN number of processes.\n");
        MPI_Finalize();
        return 1;
    }

    data = (int *)malloc(N * sizeof(int));

    // --- PATTERN II LOGIC ---
    int half = size / 2;

    if (rank < half) {
        // First Half (Senders)
        int partner = rank + half;

        for (int i = 0; i < N; i++) data[i] = rank * 10 + i;

        printf("Rank %d (First Half): Sending to Rank %d\n", rank, partner);
        MPI_Send(data, N, MPI_INT, partner, 0, MPI_COMM_WORLD);
    } else {
        // Second Half (Receivers)
        int partner = rank - half;

        MPI_Recv(data, N, MPI_INT, partner, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        printf("Rank %d (Second Half): Received from Rank %d (First element: %d)\n",
            rank, partner, data[0]);
    }
    // -----

    free(data);
    MPI_Finalize();
    return 0;
}
```

```
stephenrajva@Stephens-MacBook-Air-4 ~ % cd ~/Downloads/HPC-Upload\ 2/mpi-patterns
```

```
cp mpi_halfshift.c mpi_halfshift_original.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cp mpi_halfshift.c mpi_halfshift_original.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano mpi_halfshift.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_halfshift.c -o mpi_halfshift
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f "/Users/stephenrajva/Downloads/SST_Stimulator/Communication_Pattern/HalfShift_experiment/halfshift_trace.csv"
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 2 ./mpi_halfshift 100000
```

```

stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pattern/HalfShift_
dquote>
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pattern/HalfShift_experiment/halfshift_trace.csv"
0,1,400000,0.000152
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % ls -l "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pattern/HalfShift_experiment"
total 8
-rw-r--r--  1 stephenrajva  staff   20 18 Jun 11:48 halfshift_trace.csv
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pattern/HalfShift_experiment/halfshift_trace.csv"
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 4 ./mpi_halfshift 100000
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pattern/HalfShift_experiment/halfshift_trace.csv"
1,3,400000,0.000188
0,2,400000,0.000179
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pattern/HalfShift_experiment/halfshift_trace.csv"
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_halfshift 100000
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat "/Users/stephenrajva/Downloads/SST_Stimulator/Communication _Pattern/HalfShift_experiment/halfshift_trace.csv"
2,6,400000,0.000145
3,7,400000,0.000332
0,4,400000,0.000338
1,5,400000,0.000179
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns %

```